

HAT-P-36b

R-0749

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_260219 · created 2026-07-28T02:45:16+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461090.9307 +/- 0.004 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.109 +/- 0.02
Transit depth (Rp/Rs)^2	1.18 +/- 0.44 [%]
Orbital Inclination (inc)	86.27 +/- 2.56
Airmass coefficient 1 (a1)	0.8732 +/- 0.0096
Airmass coefficient 2 (a2)	0.126 +/- 0.085
Scatter in the residuals of the lightcurve fit is	1.11 %
Best Comparison Star	#1 - [72, 52]
Optimal Aperture	3.82
Optimal Annulus	4.97
Transit Duration (day)	0.0916 +/- 0.0055

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.109 ± 0.02 vs 0.126 (deviation 0.85σ)
Duration fit vs published	0.0916 d vs 0.0925 d (deviation 0.16σ)
Mid-time O–C	5.6 min
Depth SNR	5.5
Residual scatter	1.11 %

Light curve

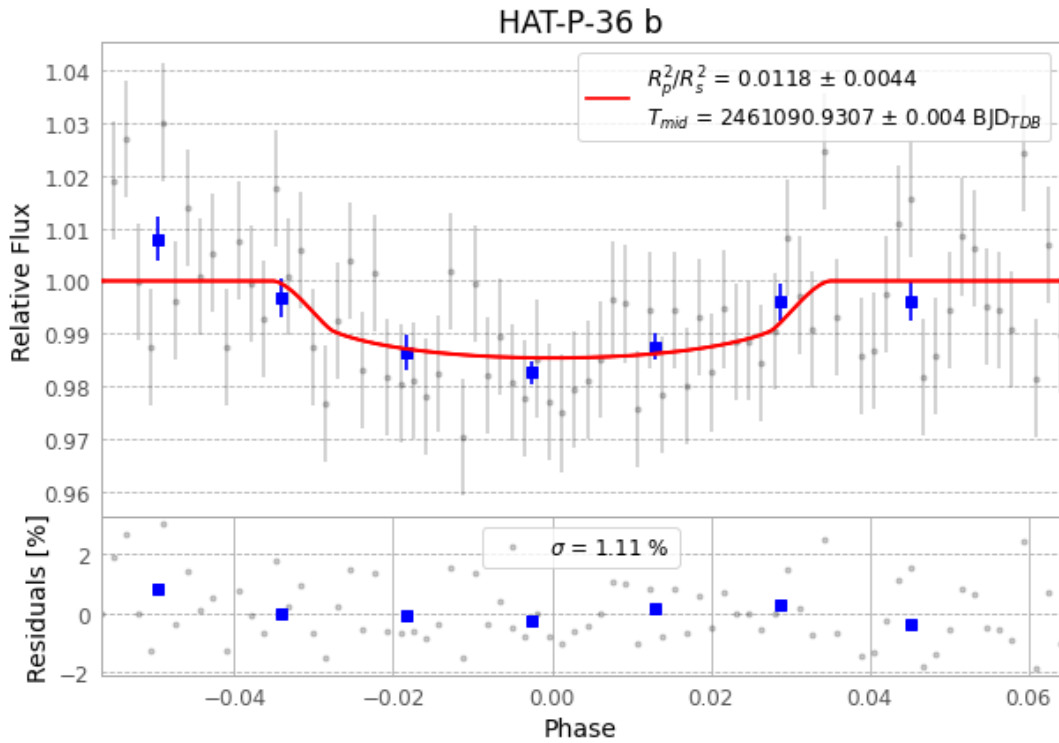


Figure 1 — FinalLightCurve_HAT-P-36 b_2026-02-19.png (SHA-256 b07adb0335cad463...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [344, 118], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 12c0d48465cd6b24...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

78 FITS frames (51 MB), HATP-36260219082415.FITS → HATP-36260219121516.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-260219.log (184a2b81b9b8...), FinalLightCurve_HAT-P-36 b_2026-02-19.png (b07adb0335ca...), FinalLightCurve_HAT-P-36 b_2026-02-19.csv (78c480cfba12...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 02:45Z.